SSU Data Dash 2026: Big Beef Burrito

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Video (YouTube, unlisted is fine): `https://youtu.be/QDTxLZXlH-8`
Code and submission files (GitHub): `https://github.com/Richardhhhg/SSU-Data-Dash`

Summary

Our team approached the analysis through an initial exploration of the data to find key trends followed by building an ensemble model. Our model yielded mdape scores of 6.75% and 3.98% on off market and list aware validation respectively. MAPE scores were at 9.18% and 5.01% for off market and list aware respectively. We found that our model was greatly limited by its inability to address the difference in trends found in the more expensive homes compared to cheaper ones.

1 Data and features

To select our features, we first made a large collection of features either from the dataset or from our own transformations and insights. Then we performed feature selection by trying subsamples of features along with backward elimination using a lightgbm model and the off market track. This resulted in 86 selected features being used across all base models in the ensemble. Both the models for off market and list aware used the same features with the exception of list aware also including the list price.

From the original dataset, we decided to keep all features except for date_listed, garage, municipality, and remarks. Features were dropped due to augmentations of the feature being made or eliminated during feature selection.

We applied keyword matching through regex to creature binary encodings for additional features about the home including: whether the house was renovated, the target buyer, and other features. Using the keyword matches, we also constructed a score for the condition of the house. Positive keyword matches about the condition increased the score while negative ones decreased it. Lastly we sought to capture the general sentiment from the remarks column by training a linear regression model with l2 regularization on the tf-idf encodings. The encodings not only included single words, but also 2 and 3 word phrases in order to capture sentiment from descriptive phrases.

We created a binary indicator for whether the home was an apartment/condo (is_apartment) and built interaction features between is_aparment, house_area, bedrooms, washrooms, and parking. This was for the model to more explicitly learn how different features vary for houses and apartments/condos.

Since we were working with time series data we also decided to include features specific for time series. We added semi annual and annual lags after noticing a trend from time series visualizations. We also included cyclical features for months with sin and cos transformations.

Lastly, there is reason to believe that homes in the same building are likely to have similar pricing schemes. To capture this we included building features capturing the sold price of earlier units in the building and price per square feet among other attributes.

2 Models

Both tracks use four gradient-boosted tree models that predict the log of the sold price, combined by a ridge regression. Boosted trees suit this tabular data, with its mixed column types, missing values and non-linear effects like location. A random forest performed far worse upon testing, so we excluded it.

The base models are XGBoost, LightGBM, CatBoost and a split LightGBM: two LightGBM models, one trained on apartments and one on houses (by is_apartment), with each home assigned to the matching model. The ridge learns from out-of-fold predictions on the six months before each prediction month, so it never sees predictions on homes the base models trained on. Hyperparameters were tuned per track with Optuna on July - November 2025, before validation.

Off-market: Without a list price, the model relies on our engineered features: building and repeat-sale history, price lags, and the remarks text score. Location and type matter most, so the split LightGBM gets the largest weight (0.37, vs. 0.27 XGBoost, 0.22 CatBoost, 0.14 LightGBM), and a stronger ridge penalty (alpha 1.93) keeps weights spread out to account for the errors from different models.

List-aware: Since off-market carries twice the accuracy points, we prioritized feature engineering for that track. List-aware uses the same features plus the list price, with separately tuned hyperparameters. The list price already captures most of a home's value, so the models mainly learn the list-to-sold gap. The ensemble leans on XGBoost (0.46), and the ridge penalty is weaker (alpha 0.37).

3 Validation

To evaluate how well our models would perform on unseen homes, we used walk-forward validation instead of a random train-test split. Before the final validation, we used July to November 2025 for time-based cross-validation and hyperparameter tuning, including Optuna optimization. This helped us select model settings using earlier data and reduce the risk of overfitting to the final validation period. For our final validation, we used December 2025 to May 2026, we trained the models using only sales completed before each prediction month. By repeating this process for each month, we were able to better reflect how the models would perform when predicting future home prices using only past information.

In addition, we took several steps to prevent data leakage throughout the validation process. Since some of our features relied on historical sales data, we rebuild them for each validation fold using the available training history, including historical price lags, market-level statistics, and past sales from the same building. We also excluded actual sale prices and other information unavailable at prediction time, including list prices for the off-market model. Finally, to reduce the risk of overfitting when combining our four models, we trained the ridge regression model using predictions on homes the models had not seen before. We also used predictions from the previous six months to train the ridge regression model before each validation month, allowing it to consider more recent market trends while still using only past information.

4 Results

We validated walk-forward: each month from December 2025 to May 2026 (10,069 sales) was predicted by a model trained only on earlier sales.

Off-market, the model reaches 6.75% MdAPE, cutting the nearest-sales baseline's 20.8% by about two thirds. Its median signed error of -0.99% shows slight underprediction, consistent with a model

Track	Homes	MdAPE % (95% CI)	MAPE % (95% CI)	Median signed error %
off-market	10,069	6.75 (6.60–6.94)	9.18 (9.00–9.37)	-0.99
list-aware	10,069	3.98 (3.88–4.07)	5.01 (4.92–5.09)	-0.14

Table 1: Walk-forward validation on the listings. MdAPE: median absolute percentage error. MAPE: mean absolute percentage error. Brackets: bootstrap 95% interval.

trained on past sales lagging a rising market. List-aware, it reaches 3.98% MdAPE, nearly halving the 7.26% error of simply predicting the list price, with almost no bias (-0.14%).

Stacking added a small gain, and a plain average scored almost as well as the ridge, so most of that benefit comes from combining different models. Error was stable across months (about 6.5–6.9% off-market, 3.7–4.1% list-aware).

Two caveats still apply though. The confidence intervals treat homes as independent, but homes sold in the same month share market swings, so the true uncertainty is wider. And validation predicts one month ahead, while the hidden test reaches up to four months ahead, so we expect higher test error.

5 Where the model fails

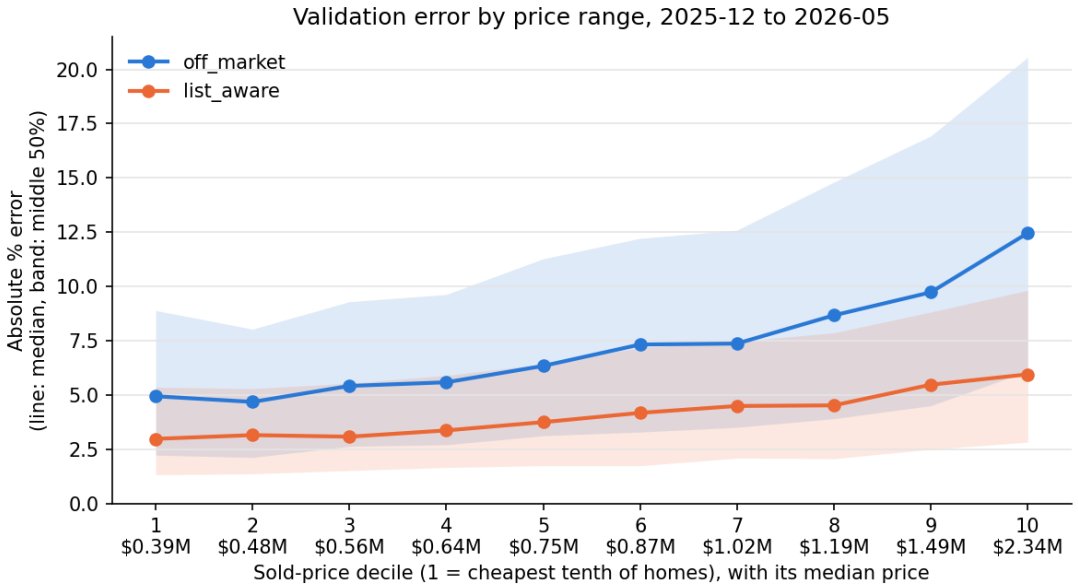


Figure 1: Absolute percentage error by sold-price decile.

Our model struggles to accurately predict prices for properties at the extreme ends of the distribution, particularly high-end properties. This is likely due to sparse training data in these price ranges, which limits the model's ability to capture underlying trends. Additionally, houses (defined as is_apartment = 0) exhibit higher error rates than apartments, which we attribute to a smaller sample size and subsequent underfitting for this property type. The more expensive data points were also almost all houses, which indicates that houses typically had a tendency to have a higher upper range for sell price compared to apartments.

We also noticed that homes that had a higher number of rooms, including bedrooms or washrooms,

had a lot higher residuals. A likely reason is sparse data at the high-room end. In the 150,000 training sales, only 0.6% had 6 bedrooms, and 4.5% had 5 or more washrooms. In each walk-forward validation month, the model has even fewer examples of those homes to learn from. Expensive, large homes may also vary more in ways room counts alone don't capture, such as lot size, condition, location, or renovation quality.

Errors for the summer also appeared to be a lot higher than everything else. There are some reasons that could factor into this. We considered possible trends that could impact the fluctuations of housing prices in summer compared to other seasons. For example, families with children still in kindergarten to highschool might prioritize moving in the summer since it's summer vacation, which wouldn't impact their children's schooling. Another reason is weather; with less precipitation, summer would be more ideal for moving trucks compared to driving through the ice and snow in winter.

6 Limitations

Our model is less reliable for properties at the high end of the price distribution, as highlighter earlier. For all of our models, they performed much better on cheaper homes compared to expensive ones when looking at the mean absolute percentage error. This suggests that expensive properties are harder to price accurately in this experiment, though the comparison is across all property types, not houses alone. One potential improvement is to model price quantiles or train specialized models for different price ranges, allowing the model to capture different patterns among lower, middle, and higher-priced homes.

Next, the feature set already includes listing date and month features, as well as six and twelve-month price lags. However, our analysis of cyclical market behavior and autocorrelation was limited. As a next step, we could analyze trends in market-level price series (for example, by property type or area) and evaluate time-series approaches such as ARIMA alongside the existing models. Any added method should be tested using walk-forward validation to avoid training on future sales.

Third is that the data had uneven coverage across years. The training data covers January 2021 through May 2026, so it contains no 2020 sales and only part of 2026. Listings began in February 2020, but the first sale was in January 2021. Sales are also unevenly distributed: 42,409 (28.3%) are from 2021, compared with 22,529–27,297 (15.0%–18.2%) in each full year from 2022 through 2025, and 2026 was a partial year. This imbalance means the model sees substantially more 2021 examples than examples from any later full year, which may affect how well it learns market conditions across time.

Our off-market and list-aware models could benefit from more tailored feature selection. The list-aware model has access to list price, which may overlap with information captured by other features such as home size and room counts. In our XGBoost model, list price had the highest reported feature-importance value. When features contain overlapping information, their individual importance can be difficult to interpret, and the model may rely heavily on list price. However, we didn't establish that this overlap harms accuracy. Future work could compare feature sets for each track and test whether the list-aware model remains accurate when list prices are unusually high or low relative to sale prices.

Reproducing our results

```
pip install -r requirements.txt
python run.py validate && python run.py report && python run.py predict && python run.py
check
```

Outside data

Source	URL	Periods covered
None		

Appendix

Additional Sources of Feature Ideas

Meta Models (2025), London House Price — LightGBM, https://www.kaggle.com/code/metamodels/london-house-price-lightgbm

Ali Soltani, Mohammad Heydari, Fatemeh Aghaei, Christopher James Pettit, Housing price prediction incorporating spatio-temporal dependency into machine learning algorithms, Cities, Volume 131, 2022, 103941, ISSN 0264-2751, https://doi.org/10.1016/j.cities.2022.103941.